



Course: Electronic Circuit Design
Program: BS (Electrical Engineering)
Duration: 60 Minutes
Paper Date: 15-Nov-2018
Section: ALL
Exam: Midterm-II

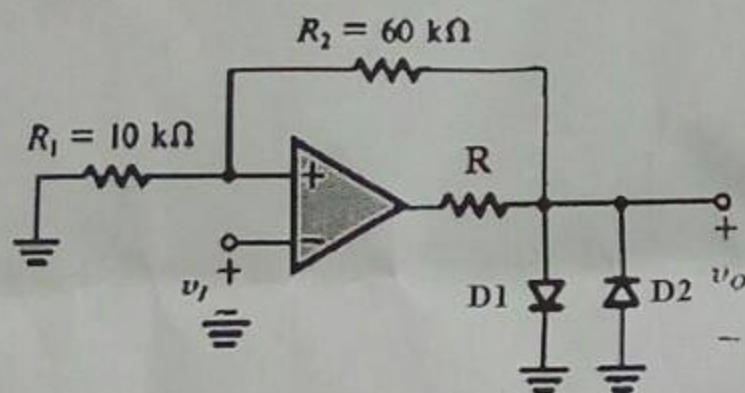
Course Code: EE224
Semester: Fall 2018
Total Marks: 30
Weight: 15%
Page(s): 1
Questions: 3

- Q1** An ideal op-amp-based inverting integrator has a voltage-gain of magnitude 100 V/V at a frequency of 1.0 kHz. Use frequency-domain based analysis techniques to find:
- The frequency at which the magnitude of the gain reduces to 1.0 V/V.
 - The value of resistor, if a $0.01 \mu\text{F}$ capacitor is used in the circuit.

[10]
CLO
03

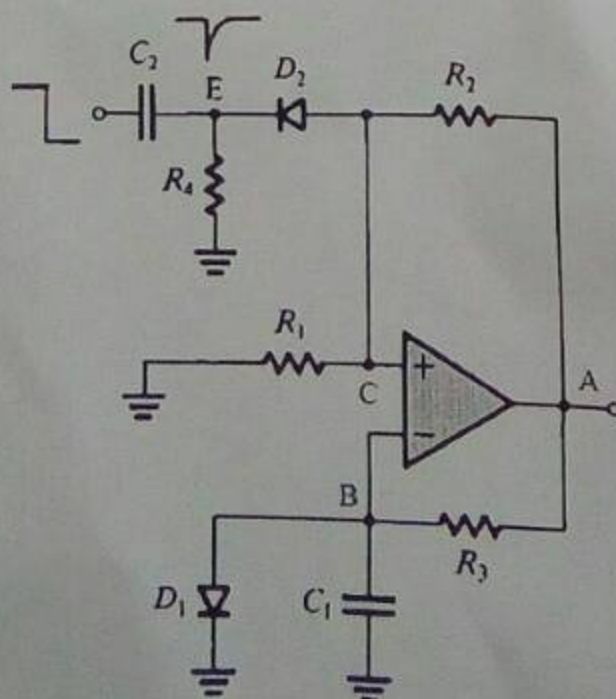
- Q2** a) For the following circuit, sketch and label the transfer characteristic v_o-v_i . The diodes are assumed to have a constant 0.7-V drop when conducting, and the immediate output of the op amp saturates at $\pm 12 \text{ V}$.
- b) Design the value of R , such that the maximum diode current in $D1$ is 1.12 mA when the op-amp output is saturated at +12 V.

[10]
CLO
04



- Q3** For the mono-stable circuit of the following figure find the value of R_3 that will result in a 200- μs output pulse for $C_1 = 0.1 \mu\text{F}$, $\beta = 0.2$, $V_D = 0.7 \text{ V}$, and $L_+ = -L_- = 15 \text{ V}$.

[10]
CLO
04



National University of Computer and Emerging Sciences, Lahore Campus



Course:	Electronic Circuit Design	Course Code:	EE224
Program:	BS(Electrical Engineering)	Semester:	Spring 2017
Duration:	60 Minutes	Total Marks:	30
Paper Date:	20-Feb-2017	Weight	15%
Section:	ALL	Page(s):	1
Exam:	Midterm-I	Questions	3

Q1 An amplifier with $R_i = 1 \text{ k}\Omega$, $G_m = 60 \text{ mA/V}$, $R_o = 100 \text{ k}\Omega$ is fed with a sinusoidal voltage source of 10 mV peak, having a source resistance of $2 \text{ k}\Omega$. It is loaded with a $0.1 \text{ k}\Omega$ resistance. Identify the nature of amplifier model. Find the output voltage. Draw the circuit schematic as well. [10]
CLO 01

$$G_1 = \frac{1 + \frac{R_o}{R_i}}{R_i}$$

Q2 a. An internally compensated open-loop op-amp has an open-loop gain of 80 dB . The corner frequency occurs at 6 Hz . Find its unity gain frequency (0 dB frequency). [13]
CLO 02

b. If the same op-amp is now connected in a closed loop to form an inverting amplifier with a closed loop gain of $G = -9 \text{ V/V}$. Find the corner frequency for this closed loop amplifier. [5]

c. Hence draw this inverting amplifier circuit diagram. The minimum resistance in the circuit should be $100 \text{ k}\Omega$. [3]

Q3 Design and construct a summing amplifier that gives the following output signal for an input signal of V_i . The resistors to be used in the circuit must be of $100 \text{ k}\Omega$. Other components must be designed accordingly. [07]
CLO 02

$$V_o = - \left(V_i + \frac{1}{2} \frac{dV_i}{dt} \right)$$

Electronic Circuit Design

Section: EE

Class Activity 1

Spring 2017

Provide a design for an AC voltmeter circuit, which is intended to function at frequencies of 10 Hz and ~~above~~ ^{below}. It should be calibrated for sine-wave input signals to provide an output of +10V for an input of 1 V rms. The input resistance should be as high as possible. To extend the bandwidth of operation, keep the gain in the AC part of the circuit ~~reasonably small as well~~ ^{unity}. The design should result in reduction of the size of the capacitor C required. The largest value of the resistor available is 1 MΩ.

$$f = 10 \text{ Hz}$$

$$V_o = 10 \text{ V}$$

$$V_i = 1 \text{ V}_{\text{rms}}$$

$$R_i \rightarrow \text{high}$$

$$\text{Gain} = 1$$

$$\frac{V_o}{V_i} = 10$$

$$V_o = 10 \text{ V}$$

$$f = \frac{1}{2\pi RC}$$

$$10 = \frac{1}{2\pi \times 10^3 \times C}$$

$$C = 1.6 \mu\text{F}$$

$$R_1 = 1 \text{ M}$$

$$R_2 = 1 \text{ M}$$

$$R_3 = 1 \text{ k}$$

$$R_4 = 10 \text{ k}$$

$$10 = 1 \left(\frac{R_4}{R_3} \right) (1)$$



Course:	Electronic Circuit Design
Program:	BS(Electrical Engineering)
Duration:	180 Minutes
Paper Date:	18-May-2017
Section:	ALL
Exam:	Final

Course Code:	EE224
Semester:	Spring 2017
Total Marks:	100
Weight	50%
Page(s):	2
Questions	5

- Q1** An amplifier for which $R_i = 1\text{ k}\Omega$, $R_o = 10\text{ k}\Omega$, and $A_o = 100\text{ A/A}$ is to be connected with a 100 mV source with a resistance of $100\text{ k}\Omega$ and a load of $1\text{ k}\Omega$. Identify the nature of amplifier model. What is the value of power gain (P_o/P_i), expressed in ratio and in dB. Draw the circuit schematic as well.

[20]
CLO
01

$$i_o/i_i = 90.9$$

$R_o - \text{upar}$

- Q2** The ideal operational amplifier in the circuit shown in Figure 1 has $v_s = 9\text{ V}$. Find v_P , v_N , and v_O ?

[20]
CLO
02

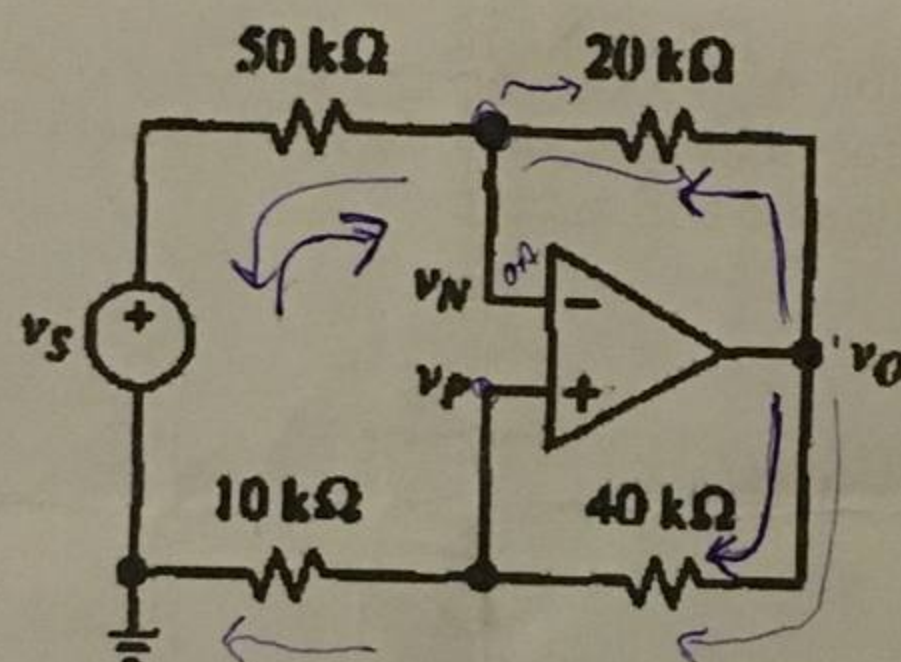


Figure 1

- Q3** The Figure 2 shows the circuit for an analog voltmeter of very high input resistance that uses an inexpensive moving-coil meter. The voltmeter measures the voltage V applied between the op amp's positive-input terminal and ground. Assuming that the moving coil produces full-scale deflection when the current passing through it is $100\text{ }\mu\text{A}$, design the value of R such that full-scale reading is obtained when V is $+10\text{ V}$.

[20]
CLO
02

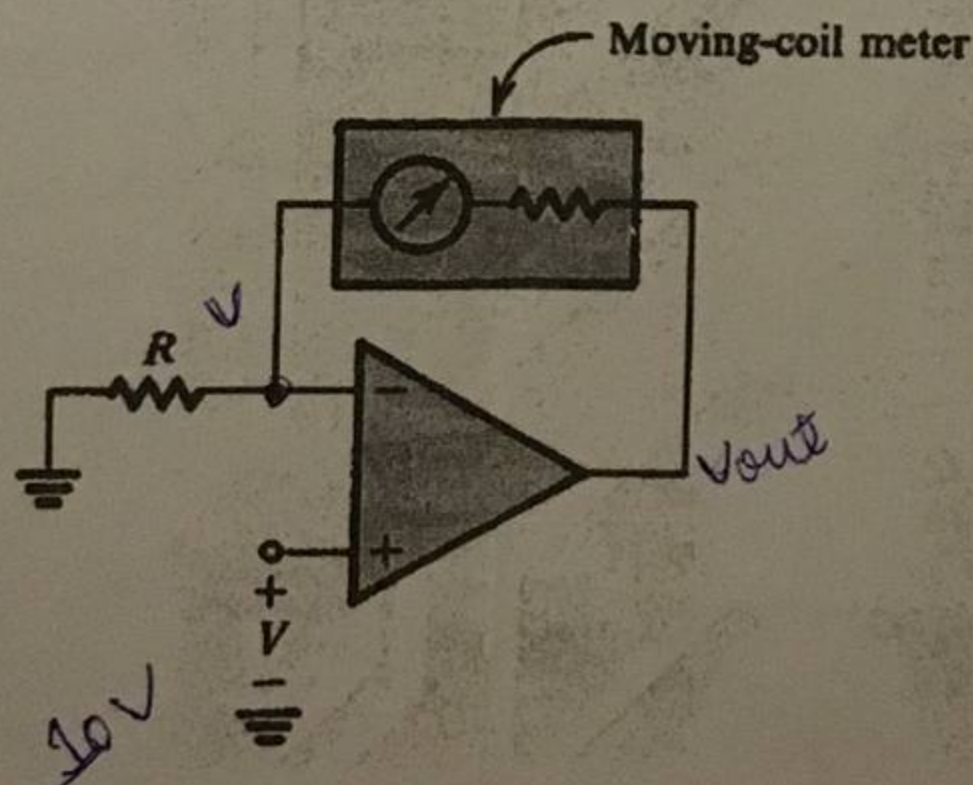


Figure 2

Q4

- (a) Draw the schematic of a monostable multivibrator using 555 timer circuit. Design the circuit for the value of external resistance (R), if an output pulse-width of 1.1-ms is to be generated using a $10\ \mu\text{F}$ capacitor.
- (b) Identify the multivibrator circuit shown in Figure 3. Design the circuit for the value of R_1 , if the frequency of oscillation is 995 Hz, $R_2 = 16\ \text{k}\Omega$, $C = 10\ \text{nF}$ and $R = 62\ \text{k}\Omega$.

[20]
CLO
04

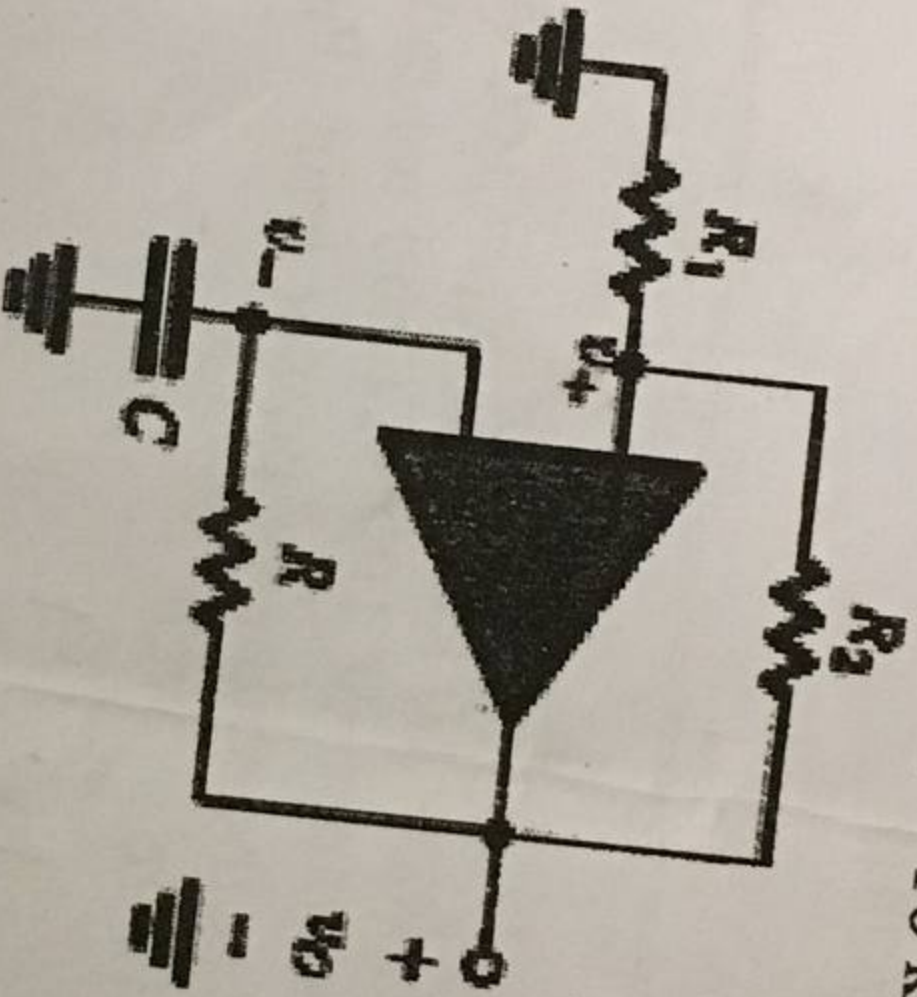


Figure 3

$$T = RC \ln(3)$$

- Q5 Draw the CMOS implementation of $Y = AB(C + D)$. Clearly show the synthesis process. Use any method for implementation. But **do NOT** invert the inputs (A, B, C and D) before applying them to the CMOS circuit.

[20]
CLO
06